

Matthew Clauson

Geometallurgy | Process Analytics | Statistical Modelling

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Profile

Geometallurgy and process analytics specialist with 10+ years' experience across mining and mineral processing. Combines geology, mineralogy, statistics and software engineering to build practical decision tools for sampling, domaining, ore control, plant troubleshooting, forecasting and uncertainty analysis. Experienced in translating complex, multi-source data into operational action and in developing technical capability across teams.

Experience

Founder & Principal Consultant | *Clauson Geometallurgy Consultancy* Sep 2026 – Present
Perth, Australia

- Developing novel statistical workflows and software for geometallurgical sampling, domaining, uncertainty quantification and predictive modelling.
- Translating research methods into reproducible R and Python tools that support testwork design, QA/QC and mine-to-mill decisions.

Senior Geometallurgist – Process Analytics | *Fortescue* Sep 2025 – Present
Perth, Australia

- Supported a cross-functional Iron Bridge troubleshooting team by integrating geology, mineralogy and plant data to identify root causes of product-grade variability.
- Built diagnostic, causal and forecasting workflows to quantify drivers of grade, throughput, recovery, energy use and product quality.
- Built reusable Python and SQL pipelines plus user-facing trial tools for structured data capture, QA/QC, reconciliation and operational decision-making.

Senior Geometallurgist – Data Science | *Fortescue* Jan 2023 – Sep 2025
Perth, Australia

- Developed a risk-analysis tool used across six major site evaluations to identify and mitigate mud-rush hazards.
- Automated QA/QC of external metallurgical laboratory datasets, removing more than 40 hours of manual processing each month.
- Applied machine learning to predictive block-model domaining during a three-month drilling campaign, improving repeatability and resource-definition decisions.

Specialist Geometallurgist – Data Science | *South32* May 2021 – Jan 2023
Perth, Australia

- Developed a site-specific geometallurgical framework linking ore characteristics to plant response and supporting throughput improvement without additional capital.
- Transferred successful GEMCO practices through structured coaching and technical training for junior staff.

Geometallurgist | *CITIC Pacific Mining* Sep 2019 – May 2021
Cape Preston, Australia

- Automated mineralogical analysis of polished blocks and grain mounts, reducing analysis time by 40%.
- Designed and implemented stereological correction workflows for magnetite ore, strengthening mineral characterisation and reporting.

Project Metallurgist | *Western Areas* Aug 2018 – Sep 2019
Cosmic Boy Concentrator, Australia

- Delivered plant-installation milestones and operator training for commissioning of a new bioleaching circuit.

Earlier experience: Project Metallurgical Technician, Western Areas (2017–2018); Metallurgical Technician / Geology Assistant, Fortescue (2017); Research Geochemist and Shift Chemist, SGS / Wolf Minerals (2016); Graduate Metallurgist, SGS Mineral Services (2015–2016).

Technical Capability

Geometallurgy: domaining, sampling and QA/QC, geostatistics, mineralogy, ore-control alignment, testwork design, mine-to-mill integration.

Statistics & modelling: Bayesian and hierarchical modelling, causal inference, experimental design, spatial statistics, time series, machine learning, uncertainty quantification.

Technology: Python, R, SQL, Snowflake, Power BI, Git, Kedro, MLflow, reproducible analytics and application development.

Processing: comminution, gravity and magnetic separation, flotation, bioleaching, process modelling and plant commissioning.

Education

Graduate Diploma of Science (Mathematics & Statistics), University of Southern Queensland – High Distinction 2025

MSc Mining Geology, Camborne School of Mines, University of Exeter 2015

BSc (Hons) Geology, University of Plymouth 2014

MSc dissertation: Digital image processing and artificial neural networks for fingerprinting a heavy-mineral-sands deposit.